For business leaders, however, there is a challenge: For all the promise, there is a counter-narrative to contend with. Skeptics say the reality doesn't match the hype. A key criticism is that agents are unreliable. Since they are built on language models (LMs), which have a meaningful error rate, they can make mistakes, fabricate information, get stuck in feedback loops, and diverge from intent. Combined with their limited memory, agents struggle particularly in tasks that have multiple steps. In one now notorious instance, an AI coding agent deleted a software company's live production database. Detractors also point to security vulnerabilities that let hackers hijack agents to steal data, manipulate systems, and spread malware. Gartner believes over 40 percent of agentic AI projects will be canceled by the end of 2027, "due to escalating costs, unclear business value, or inadequate risk controls."

So what is going on here—are agents the future of enterprise, or is it all just hot air? The truth is not so binary. Enterprises are already deriving value from AI agents, but these are not the general-purpose, autonomous "digital co-workers" that the industry rhetoric sometimes implies. Successful deployments tend to involve more constrained systems, human supervision, and specific, contained use cases: A practical approach that aims to mitigate current limitations. Realizing the broader transformational promise will require overcoming a number of technical hurdles.

Business leaders should seriously explore agents as they stand today, taking a clear-eyed view of what they are and where they can offer value across the enterprise. At the same time, they should track developments and be prepared to act fast if the field makes further advances and the dream of autonomous super-assistants starts to materialize.

